BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: DIK-HSSC-XII-2021 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 20 | **Paper Code:** DIK-HSSC-XII-2021 (A)

Q1. Choose the correct option i.e. A,B,C,D and fill it up in the correct bubble in the given MCQs sheet.

Qi. If f(x) = {3x+2 for x≤1; x^2-1 for x≥1} then f(−1) = _____

(A) 0
(B) 1
(C) −2
(D) −1

Qii. lim it(A→2)((x^2-4)/(x−2)) = _____

(A) 0
(B) 4
(C) 2
(D) Undefined [illegible in source scan]

Qiii. If f(x) = x^2 then If f'(x) = _____

(A) $x^3/3 + c$
(B) 2x
(C) $2x^2$
(D) $3x^3$

Qiv. If y = cos(x + π/2) then dy/dx = _____

(A) −sin(x + π/2)
(B) sin(x + π/2)
(C) π/2 sin(x + π/2)
(D) Zero

Qv. Second derivative of f(x) = $8x^3 - 9x^2 + 6x + 4$ is _____

(A) 48x + 18
(B) 48x − 18
(C) $24x^2 - 18x + 6$
(D) Zero

Qvi. Domain of the vector function $2ti - 3tj + t^{-1}k$

(A) {0}
(B) R
(C) R − {1}
(D) None [illegible in source scan]

Qvii. f $dx/\sqrt{(a^2-x^2)}$ = _____

(A) $1/a \sin^{-1} x/a$

(B) $\tan^{-1} x/a$

(C) $\sin^{-1} x/a$

(D) $1/a \tan^{-1} x/a$

Qviii. f_3^4 5dx = _____

(A) −5

(B) 5

(C) 35

(D) 17

Qix. Distance of the point Q(0,3) from the origin is _____

(A) 3

(B) Zero

(C) 9

(D) −3

Qx. Slope of the line 4x + 7y = 1 is _____

(A) 4/7

(B) 7/4

(C) −4/7

(D) −7/4

Qxi. Equation of the circle having center (−a, −b) and radius (a + b) is _____

(A) $(x-a)^2+(y-b)^2 = (a+b)^2$

(B) $x^2 + y^2 + a^2 + b^2$ [illegible in source scan]

(C) $(x+a)^2+(y+b)^2 = a+b$

(D) $(x+a)^2+(y+b)^2 = (a+b)^2$ [illegible in source scan]

Qxii. The equation $x^2 + y^2 + 2gx + 2fy + c = 0$ represents _____

(A) Ellipse

(B) Circle

(C) Parabola

(D) Hyperbola [illegible in source scan]

Qxiii. The parabola $x_2 = -8y$ opens _____

(A) Downward

(B) Upward

(C) To the left

(D) Right

Qxiv. Coordinates of the end points of the major axis of the ellipse $x^2/4 + y^2/9 = 1$ are_____

(A) (±2, 0)

(B) (0, ±+3)

(C) (2,3)

(D) None

Qxv. If dy/dx = cos x then y = _____

(A) sinx+c

(B) −sinx+c

(C) Cos x+c

(D) None

Qxvi. The differential equation $y' = (xy-y^2)/x^2$ = is _____

(A) Linear

(B) Not homogeneous [illegible in source scan]

(C) Homogenous

(D) None of these

Qxvii. lim it((x,y)→(0,0))(x^3+y^3+4x [illegible in source scan]) = _____

(A) −4 (B) 4

(C) 0 (D) 6

Qxviii. Sin y is homogeneous function of degree _____

(A) Zero (B) 1

(C) 3 (D) None

Qxix. f(10 + a)dx = _____

(A) 10x + c (B) ax + c

(C) (10 + a)x + c (D) Zero

Qxx. If $10^x = 2$ then x = _____

(A) Log 2 (B) Log 10

(C) 2 (D) 10

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 50

Q2. Attempt any TEN parts. Each part carries equal marks.

i. Evaluate: limit(x→0) [(1/(x+3)) − 1/3]/x

ii. Differentiate y = [(x^2+tan x)/(3x+2 tan x)]

iii. Find dy/dx when y = xe^(lnx)

iv. Find the equation of tangent line to the curve y = √(x + 1) at x = 3

v. If F(t) = (i + e^tj + t2k) and G(t) = ($3t^2$i + e^{-t}j − 2tk) are two vector functions then find d/dt(F.G)

vi. Evaluate f $\sin^4$x cosx dx

vii. Find the value of $\int_1^2(2x^{-2} - 3)dx$

viii. Find the co-ordinate of the point c(x,y) which divides the line segment joining the point A(1,2, B(3,4) in the ratio 5:7

ix. Find the center and radius of the circle $x^2 + y^2 - 4/3\ x + 4/5\ y + 19/45 = 0$

x. Write an equation of hyperbola having foci (5,0), −5,0) and eccentricity is 5/3

xi. Show that y = x − xlnx is the solution of x dy/dx + x − y = 0

xii. Find partial derivative f_y for the function of f(x,y) = $xy^3\tan^{-1}$y

xiii. Use implicit rule to find the second derivatives of $x^2 + y^2 = r^2$

SECTION – C

Maximum Marks: 30

Note: Note: Attempt any Three Questions. All questions carry equal marks:

Q 3 a) If $f(x) = 4x + 1$ and $g(x) = 2x^3 + 5x$ then find $g(f(x))$. Also state why its inverse does not exist?

b) Show that the lines $x + 4y + 3 = 0$, $5x - 4y - 5 = 0$ and $2x + 2y + 1 = 0$ are concurrent. Also find the point of concurrency.

Q. 4 a) Evaluate $\int \cos^4 x / \text{cosec } x \, dx$

b) Show that the first four terms in the Taylor series expansion of $f(x) = x + e^x$ about $x = 1$ are $(1 + e)x + e[(x-1)^2/2! + (x-1)^3/3! + (x-1)^4/4!]$

Q. 5 a) Find f'(p) when $f(p) = (2p+3)(4p-1)/(3n+2)$

(b) Find the equation of the circle passing through the pint (0,0), (0,3) and the line $4x - 5y = 0$ is tangent to it at (0,0).

Q 6. (a) For what value of 'n' the line $x - y + n$ will touch the parabola $y2 = 8x$

(b) Verify Euler's Theorem for $z = f(x,y) = ax^2 + 2hxy + by^2$